

What is a galaxy? That's a difficult question to answer Figuring out what really counts as a galaxy could shake up astrophysics, cosmology and particle physics, says **Chanda Prescod-Weinstein**

WHEN is a collection of stars just a collection of stars and when is it a galaxy? This sounds like a nerdy joke of some kind, but it is an actual scientific question the astronomy community is struggling over.

Those of us who aren't professional stargazers are used to thinking of galaxies as complete objects. When I use telescopes in my backyard to look at galaxies, I see spirals emitting the bright light of billions of stars. From my own research as a theoretical particle cosmologist, I know these galaxies are surrounded by an invisible halo of dark matter. From my graduate training, I also know not all galaxies have a spiral shape.

This makes what a galaxy is feel very straightforward, but our understanding of them is a work in progress. While it is easy to categorise something with a clear spiral structure and billions of stars, what about something that looks spheroidal with millions of stars? What I have described is the basic definition of a globular

Globular clusters are collections of stars bound together inside galaxies

cluster. These are collections of between tens of thousands and millions of stars gravitationally bound in a formation just a few light years across. Importantly, they live inside galaxies.

The fact that globular clusters are only found inside galaxies would suggest they are a distinct cosmic phenomenon from galaxies themselves. Besides, globular clusters are compact, while galaxies are more spread out across space – this is true even for dwarf spheroidal galaxies that are gravitationally bound to the Milky Way. We also now know that dwarf spheroidals are contained in their own dark matter halos, while globular clusters aren't. For some, this is what separates the two: to be a galaxy is to be full of dark matter.

Until 2005, this boundary seemed to work, but then the Sloan Digital Sky Survey (SDSS) released its first dataset. The SDSS, tasked with scanning over a quarter of the night sky, was cataloguing a plethora of never-before-seen objects. Within that data, astronomers found observations of very faint, hard-to-see collections of stars near

the Milky Way, which challenged the idea that it is easy to distinguish between globular clusters and galaxies.

In some cases, follow-up observations made clear the object in question was indeed a galaxy, chock full of dark matter. That work continues and isn't always straightforward. Their inherent faintness makes these collections of stars an observational challenge.

Recent sky surveys have muddied the waters further by revealing a population of even more faint objects, and we now know better than to presume they aren't galaxies. Moreover, we don't all agree on what is what: for example, a 2023 paper from Simon Smith at the University of Victoria in Canada and his team declared the discovery of the galaxy Ursa Major III, which they called "the least luminous known satellite of the Milky Way". Though the authors face a battle making their case because observations count 60 stars in the galaxy! Not 60 million or 60 billion. Just 60.

Last year, a research team claimed that if Ursa Major III is a galaxy, then it can be used to rule out a class of dark matter models. Figuring out whether Ursa Major III and other compact, ultra-faint Milky Way satellites like it are galaxies thus has the potential to shake up astrophysics, cosmology and particle physics.

Recently, William Cerny at Yale University and his colleagues published a report containing the first extensive investigation of a large group of these objects. Their conclusion? They are a mix of types, but more observations are needed. I have nothing certain to tell you about what we know, which is an exciting place to land. Our current position is mid-research excitement, the part where we stand at the edge of what we know and push past it. ■



Chanda Prescod-Weinstein is an associate professor of physics and astronomy at the University of New Hampshire. She is the author of *The Disordered Cosmos* and forthcoming book *The Edge of Space-Time: Particles, poetry, and the cosmic dream boogie*

Chanda's week

What I'm reading

Poets, especially Cortney Lamar Charleston's collection It's Important I Remember and Camonghne Felix's manifesto Let the Poets Govern.

What I'm watching

Too much Alfred Hitchcock!

What I'm working on

I'm getting ready for the US launch of my book The Edge of Space-Time on 7 April!



NASA, ESA, P. GOUDFROOIJ, GLADYS KOBER